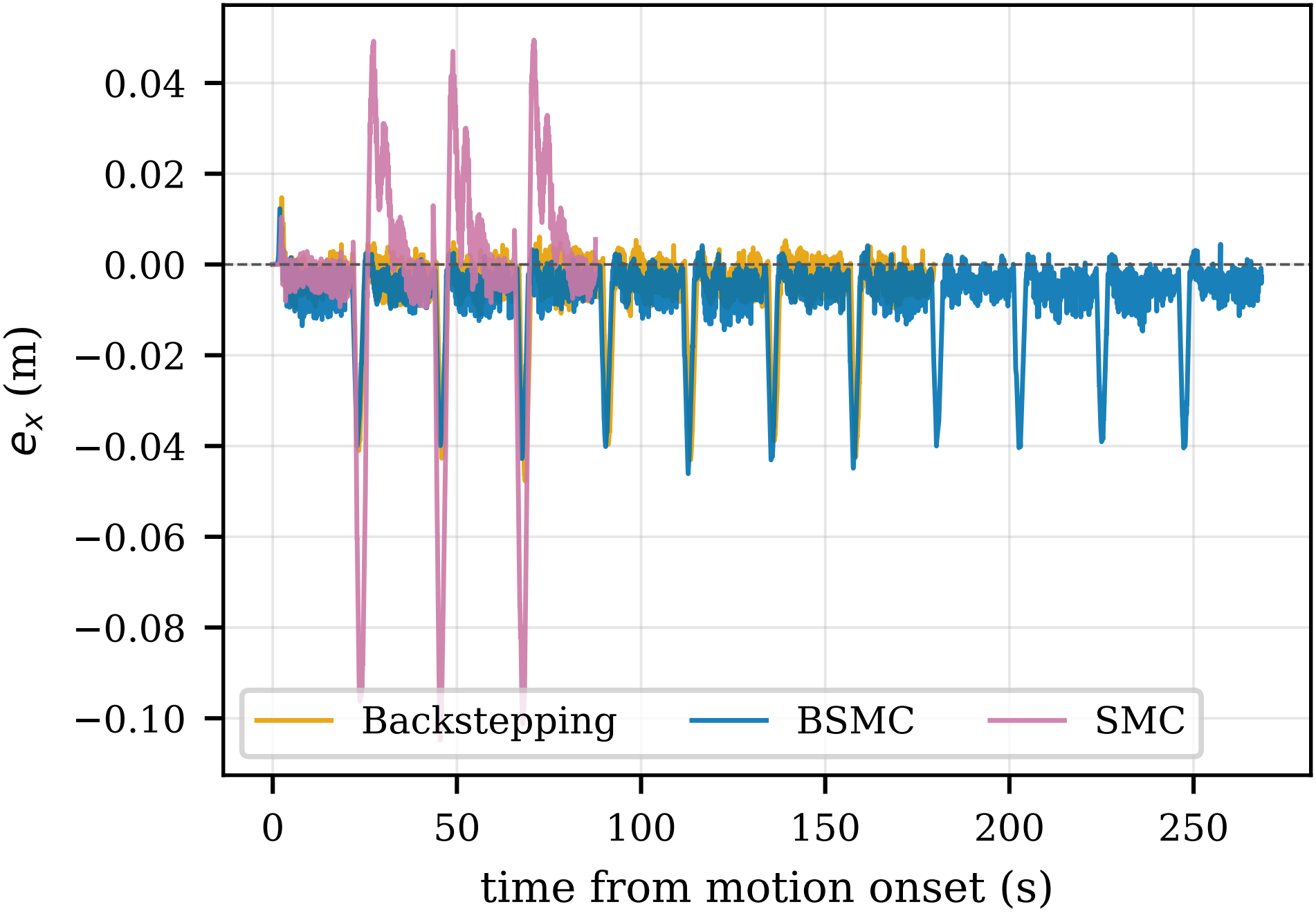

0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
$e_x$ (m)
Backstepping
BSMC
SMC
0
50
100
150
200
250
time from motion onset (s)